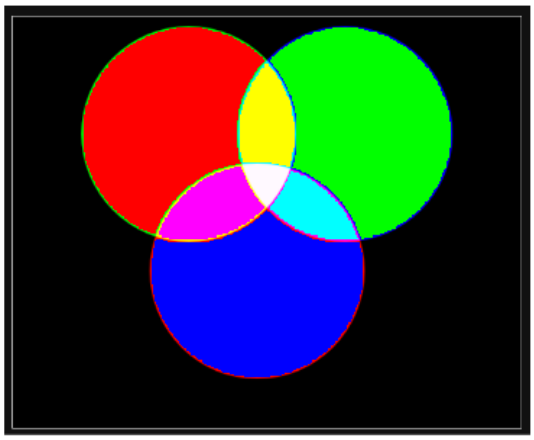
# Color Mixing with Light vs  Pigments

This handout can be used along with the Color Mixing App.  The app has two modes: one for color mixing with light and the other with pigments.  Both of these topics are covered in this document.  First color mixing of light will be briefly covered; and, then color mixing with pigments will be covered in more detail.  The idea about the two ways of mixing light is that color mixing with **light**  and with **pigments** differ because they follow two opposite systems of color creation.

## *First, Color Mixing With Light*



*Additive Color Mixing Diagram*
***figure 1***

Mixing with white light, uses **red, green, and blue (RGB) lights** as ***primary colors***.  And mixing with pigments uses cyan, yellow, and magenta (CYM)  as ***primary pigments***.  First, RGB color mixing will be covered.

Imagine that you have three flashlights, each shining one of three colors of light: red, green, and blue.  You shine them on a blackboard so that they form the pattern shown in f*igure 1*.  This kind of pattern is called a Venn Diagram.  Some parts of the diagram show only where the red light hits the board and doesn't mix with any of the other colors.  The same is true for only where the green and blue hit the board and don't mix with other colors.  The yellow, cyan, and magenta are called secondary colors. Secondary colors result from the mixing of primary colors. The white part of the diagram results from mixing all three RGB primary colors.

*As seen in figure 1*,

- when red light and green light mix, but not blue, yellow is produced.
- when green light and blue light mix, but not red, cyan is produced.
- when blue light and red light mix, but not green, magenta is produced.
- when red, green, and blue mix, the result is white light.

This is how screens, projectors, and LEDs create color.

Consider the most important reason that the color mixing of light is important. Your eyes actually have tiny sensors called *cones* that are specifically tuned to different shades of red, green, and blue. For example, when you look at a yellow emoji on a phone screen, there isn't actually any "yellow" light there. Instead, your screen is just blasting your eyes with a bit of red and a bit of green that are right next to each other, mixing together and "tricking" your brain into seeing yellow! This is very similar to what is happening in *figure 1*, where we see red light mixing with green light to produce yellow light.

## RGB (Light) Exploratory Questions and Answers

### 1. Observation from the diagram in *figure 1*.

**Question:**
In *figure 1,* what colors appear where two circles of light overlap? Why do you think these particular colors appear instead of others?

**Answer:**
Red and green make yellow, green and blue make cyan, and blue and red make magenta. These colors appear because the wavelengths of the overlapping lights combine and stimulate the eye's red, green, and blue cone cells in specific proportions.

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### 2. Reasoning about yellow light

**Question:**
If a screen can produce yellow by mixing red and green light, why do many art classes teach that yellow is a primary color?

**Answer:**
Art classes usually teach pigment mixing rather than light mixing. Pigments follow a subtractive system (CMY), where yellow is a primary color, while light follows an additive system (RGB).

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### 3. Thinking about the eye

**Question:**
Your eyes have cones that respond most strongly to red, green, and blue wavelengths. How might this influence the way electronic screens are designed?

**Answer:**
Screens use red, green, and blue pixels because those three colors can stimulate the cone cells in different combinations, allowing the screen to create the full range of colors humans can see.

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**4. Testing the idea**

**Question:**
Suppose a screen pixel emits only red and green light but no blue. What color will your brain perceive?

**Answer:**
Yellow, because the red and green cones in the eye are stimulated together while the blue cones are not.

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**5. Connecting to technology**

**Question:**
If you look closely at a television or phone screen with a magnifying glass, what tiny colored elements might you see?

**Answer:**
You would see tiny red, green, and blue subpixels that combine to produce the colors on the screen.

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**6. White light question**

**Question:**
Figure 1 shows white light produced when red, green, and blue mix. If one of these colors were removed, what would happen?

**Answer:**
The result would no longer be white. The color would shift toward the remaining two colors, such as yellow (red + green), cyan (green + blue), or magenta (red + blue).

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**7. Intensity vs. presence**

**Question:**
What would happen if red light were very bright, green light were dim, and blue light were absent?

**Answer:**
The mixture would appear reddish-orange or yellowish because red dominates the mixture while a small amount of green shifts the color away from pure red.

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**8. Comparing systems**

**Question:**
Why do screens use RGB mixing while paints and inks use a different system (CMY)?

**Answer:**
Screens produce color by emitting light, which follows additive mixing. Paints and inks create color by absorbing and reflecting light, which follows subtractive mixing.

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**9. Biology connection**

**Question:**
If humans had four different types of color-sensitive cones instead of three, how might screens need to change?

**Answer:**
Screens might need four primary light colors instead of three in order to reproduce the full range of colors that the eye could detect.

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**10. Prediction**

**Question:**
If three projectors—one red, one green, and one blue—were pointed at the same white wall and perfectly overlapped, what color would appear at the center?

**Answer:**
White light, because all three primary light colors are combined.

***Before going on:***

***A note about white light.*** As can be seen below, white light is a mix of red, green, and blue lights. Therefore, we will often use the following symbol to depict white light.

White Light

## *Mixing with pigments*

### *What is a Pigment*

***Pigments*** are substances - like ink, crayons, and paint - that absorb some colors of light and reflect others. There are three primary pigments *cyan, yellow, and magenta.*

White light is composed of the primary colors r = red, b = blue, and g =green light. A ***primary pigmen***t is a pigment that can absorb one of the primary colors and reflect back the other two that are not absorbed.

Venn diagrams can also be used to help in determining how pigments work. Consider *figures 2a* and *2b*. *Figure 2a* shows the **RGB color mixing** diagram that you worked with earlier. *Figure 2B* shows the **CYM pigment mixing diagram** that you will soon use, C is for cyan pigment, Y for yellow pigment, and M for magenta pigment. C, Y, and M are called the ***primary pigments***.

There are three primary RGB light colors and three primary CYM pigments. Here is a simple rule, a single primary pigment absorbs only one primary color. So for example, what does cyan absorb? Note in the following diagram. It looks like the red only part on the left has been replaced by cyan only on the right. This is a way to remember that cyan absorbs red. What light colors do magenta and yellow absorb?

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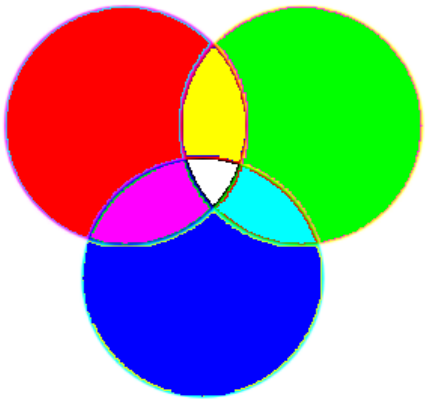


*figure 2a*

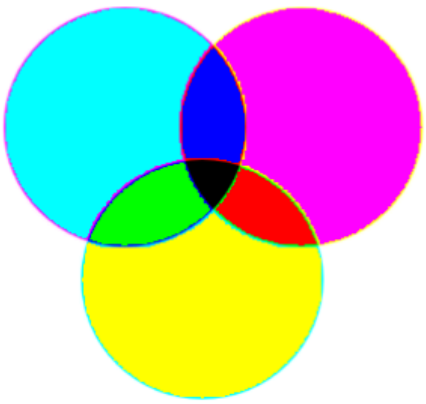


*figure 2b*

As in *figure 2a*, it might be tempting to think that *figure 2b* results from shining cyan, magenta, and yellow flashlights on a wall. But this is not the case. Even though they are both Venn Diagrams, they represent **two completely different physical processes**.

Let's try to explain by first stating some properties of how pigment mixing works. Following is the same CYM Venn Diagram, but with the different colored sections numbered. Refer to this diagram when reading the properties and rules coming up.

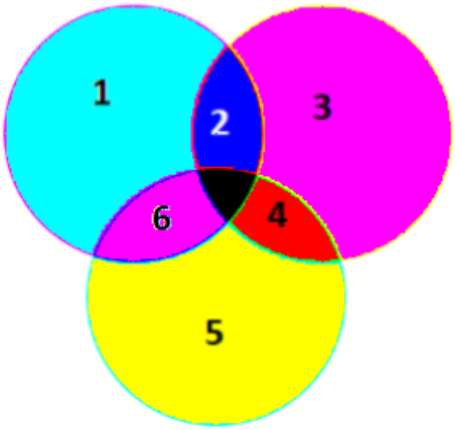
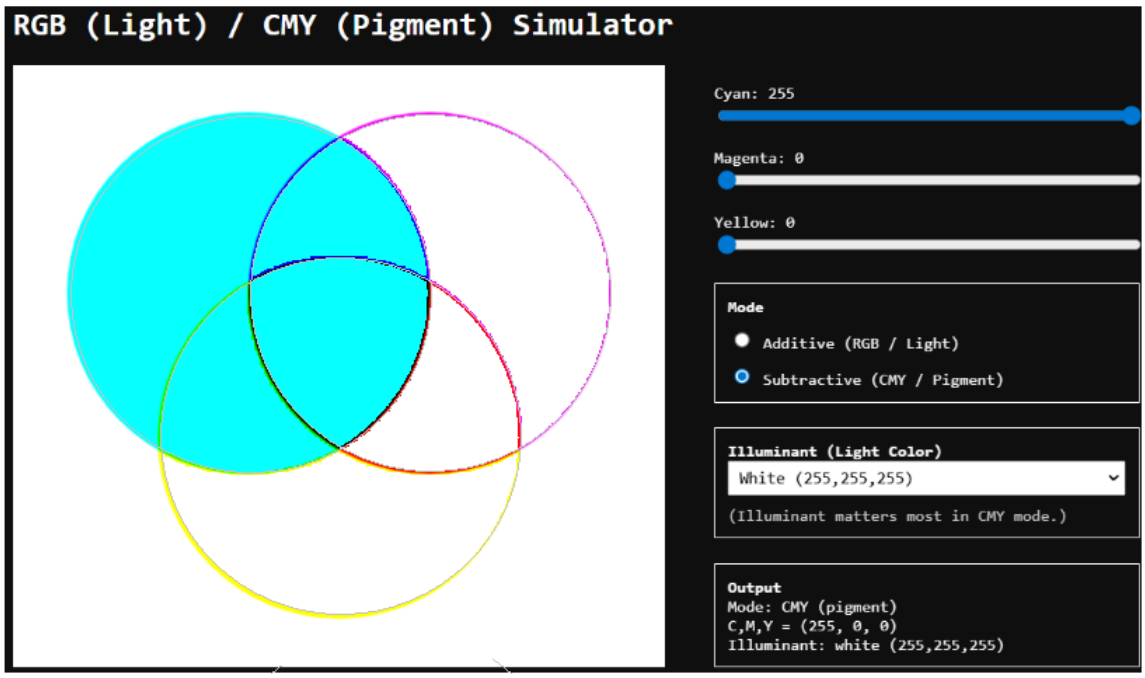



*figure 3*

Note that we are not covering the posterboard with circles of different pigments. The diagram is a key/tool that shows how pigments act when used alone or when mixed together to cover the entire surface of the posterboard.

Next let's make some observations when white (RGB) illuminates the surface of the posterboard. The following numbers aren't in order. This is OK. Also, each case will be illustrated using the *Mixing Colors Web APP,* when it is in *Subtractive Color* mode.

*Region 1:* This is the case where the entire surface of the posterboard only contains cyan pigment. So the R in the RGB is absorbed and GB is reflected, which combines to form Cyan. The color of the circle represents the color coming from all of the posterboard. The circle is there to remind us that it is part of a Venn Diagram.




*Cyan slider all the way to the right. Only Cyan pigment covers the surface.*
***figure 4***

In *figure 4,* we

*Magenta and Yellow sliders all the way to the left. (None of these two pigments are showing, just enough of their circumferences to remind us that we are working with a Venn Diagram.)*

*Mode: subtractive.*

*Illuminate: RGB white light.*

Following is what the fully illuminated posterboard would look like. The color of the cyan circle in *figure 4* tells us this.

*Posterboard - Fully Illuminated Cyan Light*
***figure 5***

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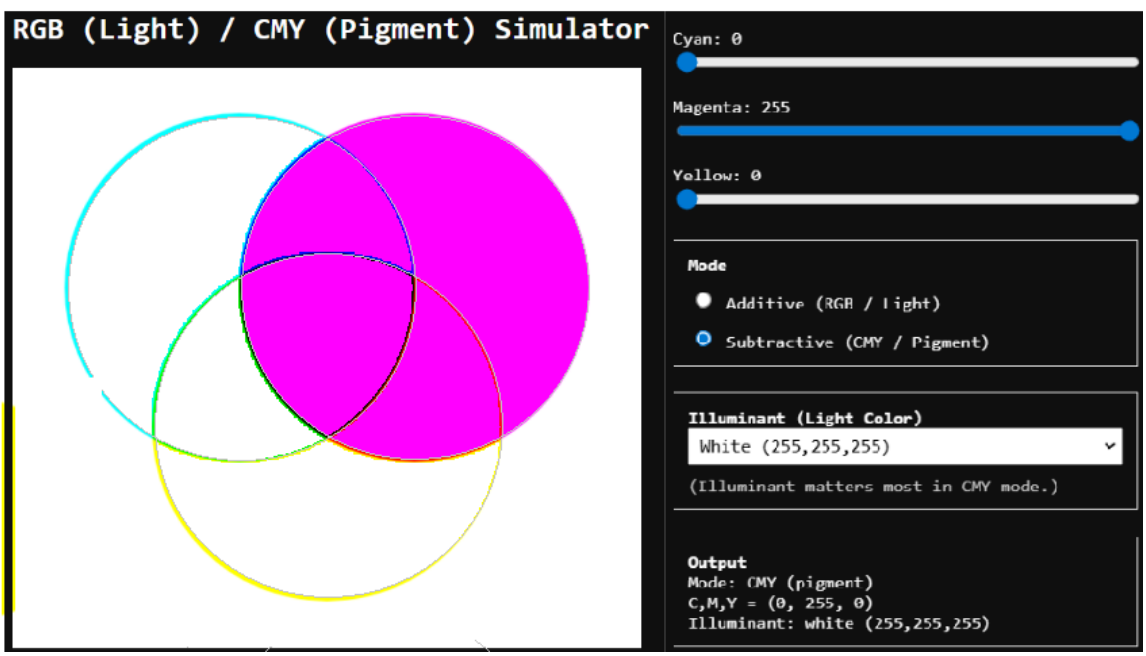
*Region 3:* This is the case where the surface of the posterboard only contains magenta pigment. So the G in the RGB is absorbed and RB is reflected, which combines to form magenta.




*Same settings as in figure 4 except for the slider values, which shows the magenta slider the only one all the way to the right.*

***figure 6***

Following is what the fully illuminated posterboard would look like. The color of the magenta circle tells us this.

*Posterboard - Fully Illuminated Magenta Light*
***figure 5***

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*Region 5:* This is the case where the surface of the posterboard only contains yellow pigment. So the B in the RGB is absorbed and RB is reflected, which combines to form yellow.

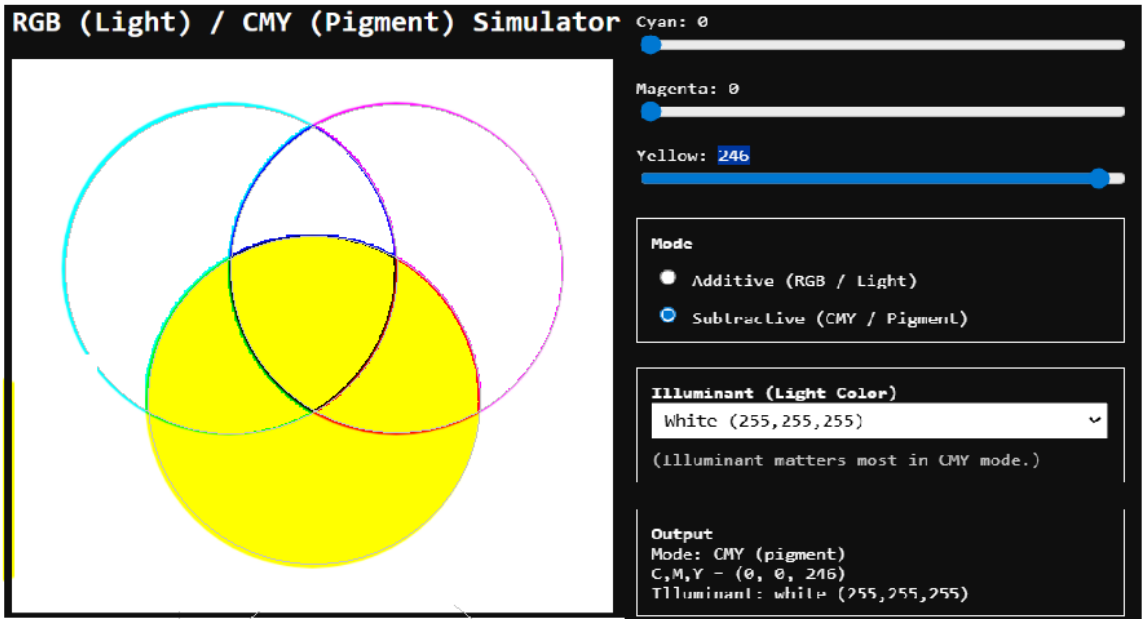


***figure 6***

Following is what the fully illuminated posterboard would look like. The color of the yellow circle in *figure 6* tells us this.

*Posterboard - Fully Illuminated With Yellow Light*
***figure 7***

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*Region 2:* This is the case where the surface contains a mixture of two pigments: cyan and magenta. In the region of overlap between the cyan and magenta circles, the R in the RGB is absorbed by the cyan pigment and the green is absorbed by the magenta pigment. This leaves just the B, which is reflected.

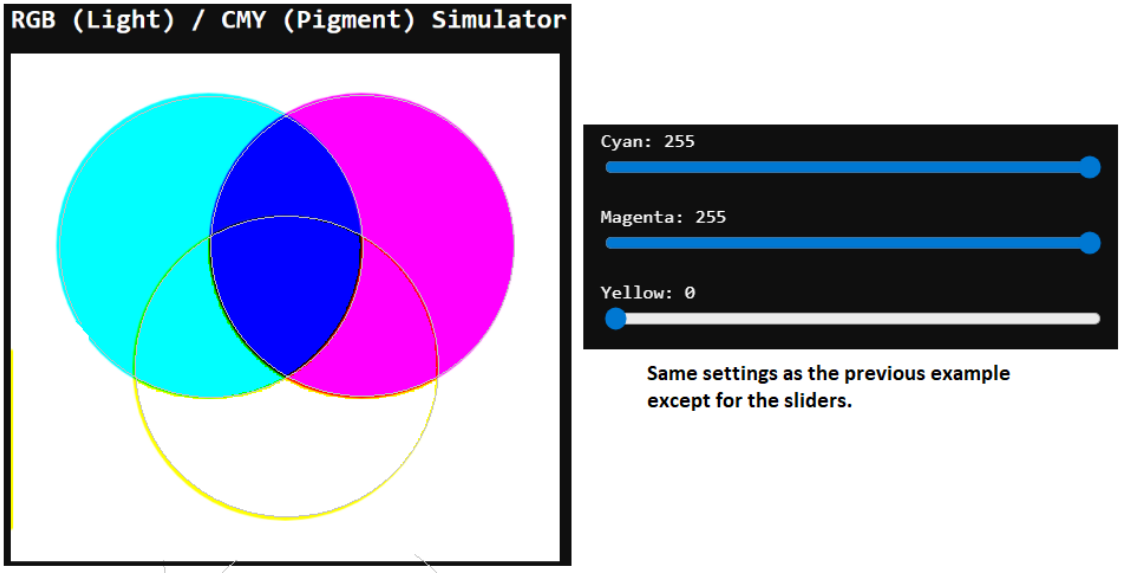
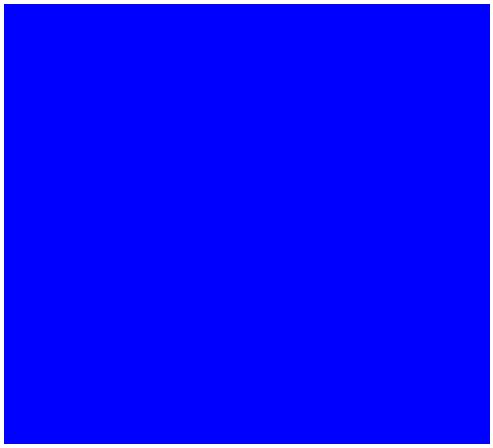



*figure 8*

Following is what the fully illuminated posterboard would look like. The color of the overlap region in *figure 6*, which is blue, tells us this.



*Posterboard - Fully Illuminated With Yellow Light*
***figure 9***

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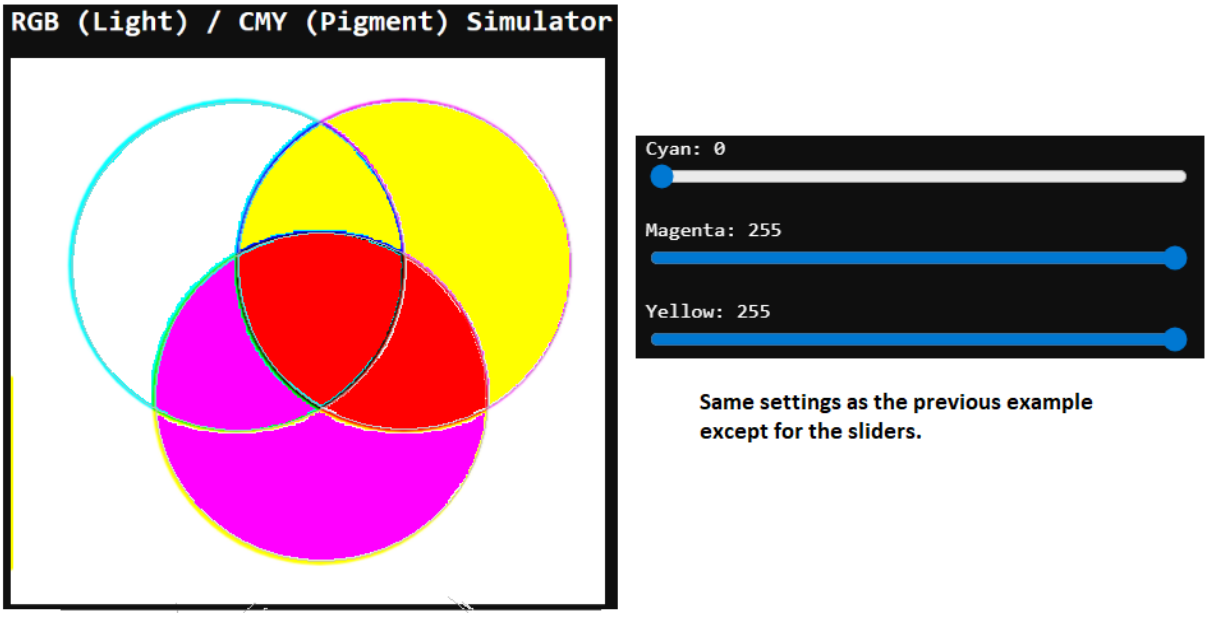
*Region 4:* This is the case where the surface contains a mixture of two pigments: yellow and magenta. In the region of overlap between the yellow and magenta circles, the B in the RGB is absorbed by the yellow pigment and the green is absorbed by the magenta pigment. This leaves just the R, which is reflected.




**Same settings as the previous example except for the sliders.**

***figure 10***

Following is what the fully illuminated posterboard would look like. The color of the overlap region in *figure 6*, which is blue, tells us this.

*Posterboard - Fully Illuminated With Red Light*
***figure 11***

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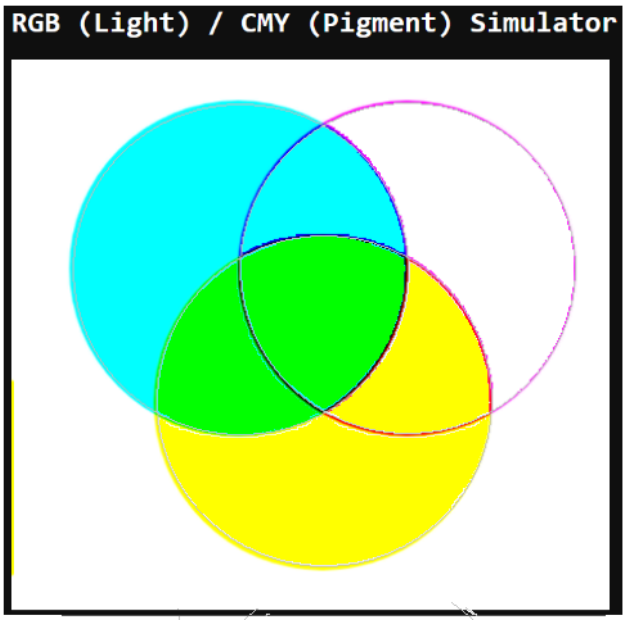
*Region 6:* This is the case where the surface contains a mixture of two pigments: cyan and yellow. So the R in the RGB is absorbed by the cyan and the B is absorbed by the yellow. This leaves G, which is reflected.



**Same settings as the previous example except for the sliders.**

***figure 12***

*Posterboard - Fully Illuminated With Green Light*
***figure 13***

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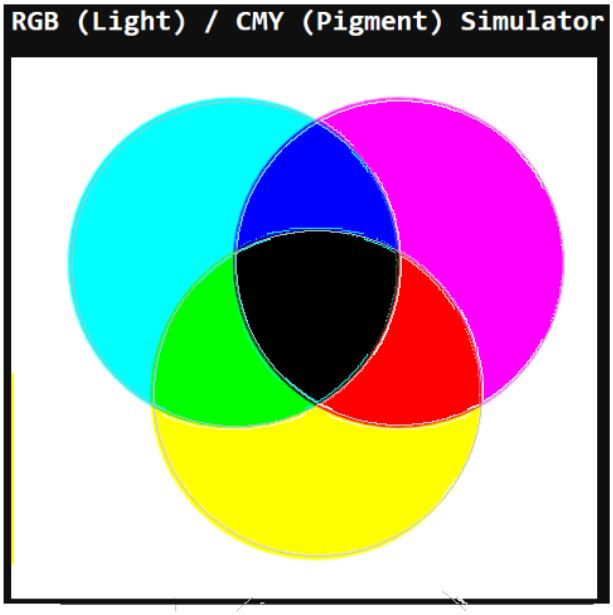
*Region 7:* This is the case where the surface contains a mixture of all three primary pigments: cyan, yellow, and magenta. So all of the RGB is absorbed and no light is reflected. The surface appears black.



**Same settings as the previous example except for the sliders.**

*Posterboard - Black, not reflection*
***figure 14***

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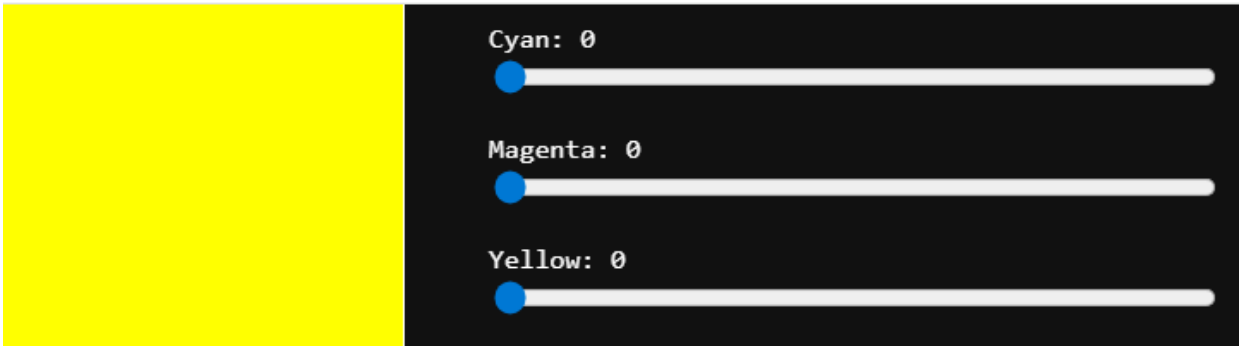
All of this is summed up in the following table.

| Pigment | Absorbs | Reflects |
|---|---|---|
| cyan | red | blue + green |
| magenta | green | red + blue |
| yellow | blue | red + green |
| cyan + yellow | red and blue | green |
| cyan + magenta | red and green | blue |
| magenta + yellow | green and blue | red |

## *Subtractive Color Mixing Using Different Color Posterboard Illuminations*

The examples that we just looked at demonstrate what happens when a surface containing no or one or more primary pigments is illuminated by white, rgb light. Next, we will look at some examples where the surface of a posterboard is illuminated by another color of light, such as yellow.

***Case 1:*** Only yellow light shines on a posterboard with now pigments.




**(R + G) - 0 = R + G = Yellow**

***figure 15***

If no pigments are used, then yellow light impinging on the poster board is reflected.

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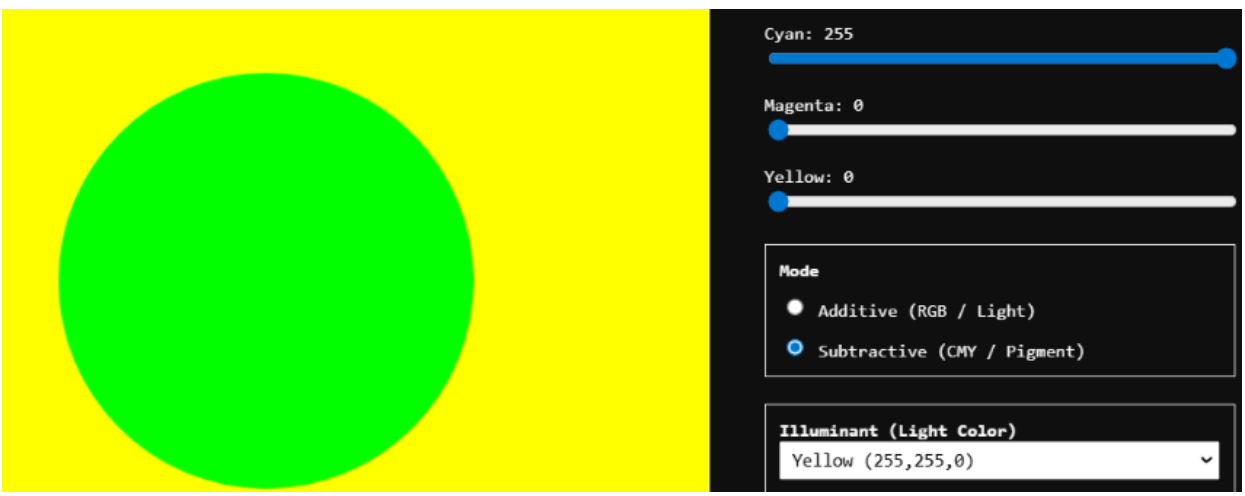
***Case 2:*** Only yellow light shines on a posterboard covered with only cyan pigment.




**(R + G) - R = G**
***Cyan absorbs the red from the yellow light and the entire posterboard reflects the green.***

***figure 16***

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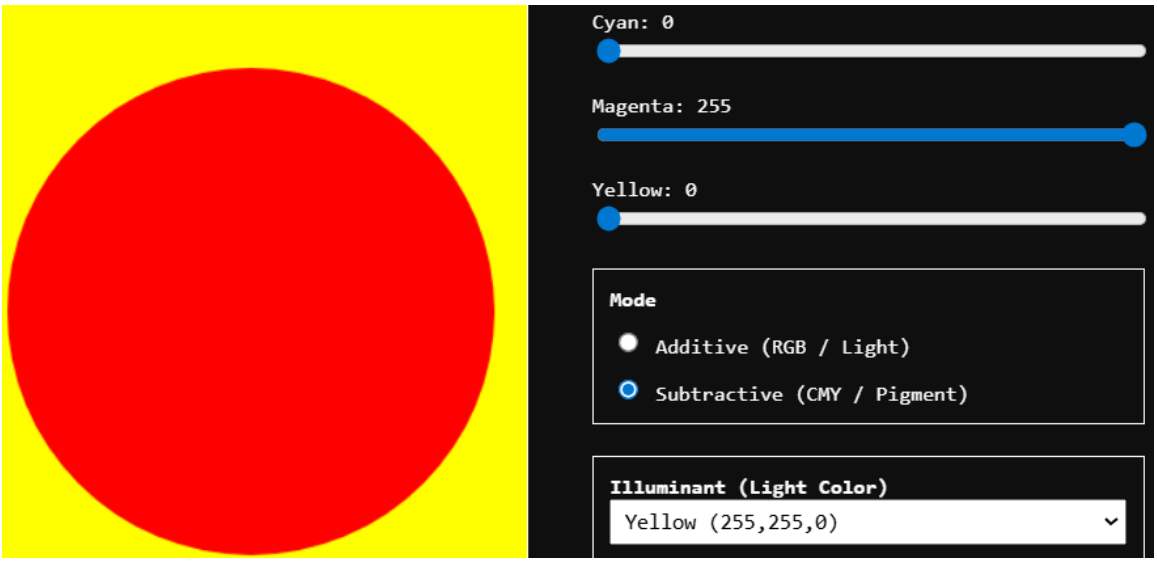
***Case 3:*** Only yellow light shines on a posterboard covered with only magenta pigment.




**(R + G) - G = R**

***Magenta light absorbs green from the yellow light and the entire posterboard reflects the red.***

***figure 17***

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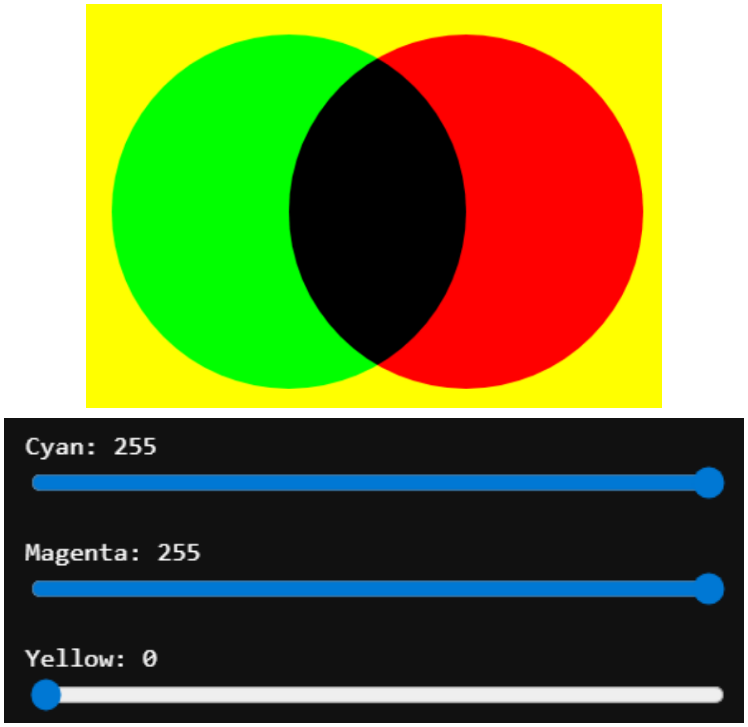
***Case 4:*** Only yellow light shines on a posterboard covered with a mixture of two pigments:
cyan and magenta.




**(R + G) - R - G = 0  (Appears Black)**

***figure 18***

*Figure 18* shows that the intersection of the two circles shows the case where the surface is composed of two pigments: cyan and magenta. The magenta part of the intersection absorbs green from the yellow light and the cyan absorbs the red.  So there is nothing left to reflect.  That's why the intersection is black.

## *CMY (Pigment) Exploratory Questions and Answers*

### 1. Observation from pigment mixing

**Question:**
When cyan and yellow pigments are mixed together, what color is usually produced?

**Answer:**
Green, because cyan absorbs red light and yellow absorbs blue light, leaving mostly green light to be reflected.

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### 2. Pigment behavior

**Question:**
Why do pigments tend to produce darker colors when more are mixed together?

**Answer:**
Each pigment absorbs certain wavelengths of light. When more pigments are added, more wavelengths are absorbed, leaving less light to be reflected.

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### *3. Cyan pigment*

**Question:**
What color of light does cyan pigment absorb, and what color does it reflect?

**Answer:**
Cyan absorbs red light and reflects green and blue light.

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### *4. Magenta pigment*

**Question:**
What color of light does magenta pigment absorb?

**Answer:**
Magenta absorbs green light and reflects red and blue light.

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### *5. Yellow pigment*

**Question:**
What color of light does yellow pigment absorb?

**Answer:**
Yellow absorbs blue light and reflects red and green light.

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### *6. Combining pigments*

**Question:**
What happens when cyan, magenta, and yellow pigments are all combined?

**Answer:**
Nearly all wavelengths of light are absorbed, so the mixture appears very dark or black.

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### *7. Printing technology*

**Question:**
Why do printers typically use cyan, magenta, yellow, and black (CMYK) inks instead of RGB?

**Answer:**
Printers work by subtracting wavelengths from white paper using pigments, which follows the subtractive color system.

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### *8. Real-world observation*

**Question:**
Why does mixing many different paints together often produce a muddy brown or black color?

**Answer:**
Because each paint absorbs different wavelengths, and mixing many paints removes most of the light that would otherwise be reflected.

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### *9. Connection to illumination*

**Question:**
How would the color of a pigment change if the light shining on it were not white?

**Answer:**
The pigment can only reflect the wavelengths present in the incoming light, so the perceived color may change.

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### *10. Comparison with light mixing*

**Question:**
Why do RGB lights become brighter when combined while CMY pigments become darker?

**Answer:**
RGB lights add wavelengths together, increasing the total light reaching the eye, while CMY pigments remove wavelengths by absorption.